

Deduplication Produces a Contamination-Correction Benchmark Accuracy Signature: Evidence from Token-Count-Matched Pythia Model Comparisons

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Abstract

Training data deduplication is widely believed to improve language model quality, yet when we compare Pythia models trained on the Pile versus its deduplicated variant at token-count-matched checkpoints, deduplication makes MMLU accuracy *significantly worse* while improving HellaSwag and ARC-Challenge. We show this divergence is not paradoxical — it is a contamination-correction signature: the direction and magnitude of each benchmark’s accuracy change under deduplication correlates positively with the benchmark’s estimated n-gram overlap with the Pile training corpus (Pearson $r = 0.632$, $p = 0.0086$ across 16 observations spanning 4 benchmarks and 4 model sizes 160M–6.9B). Benchmarks most contaminated in the original corpus show the largest accuracy reductions under deduplication, while low-contamination benchmarks are unaffected or improved. We additionally demonstrate that token-count matching — not step-matching — is the correct confound-control methodology for such comparisons, recovering a $\Delta r = 0.093$ stronger contamination signal than step-matched analyses. Our findings reframe deduplication’s benchmark effects from a uniform quality improvement to a selective contamination correction, with direct implications for how practitioners should interpret benchmark regressions after corpus deduplication.

1. Introduction

Training data deduplication is widely prescribed as a way to improve language model quality — yet when we evaluate Pythia models trained on the Pile versus its deduplicated variant at precisely token-count-matched checkpoints, dedu-

plication makes MMLU accuracy scores significantly *worse* ($p = 0.0114$, Bonferroni-corrected across four benchmarks). Meanwhile, HellaSwag and ARC-Challenge scores improve. The deduplication algorithm removed the same content; the benchmarks responded in opposite directions. What is happening?

The standard explanation — that deduplication produces a uniformly better, cleaner model — cannot account for this divergence. Instead, we argue that deduplication is better understood as a **contamination-correction mechanism**: it selectively removes training-corpus content that overlapped with benchmark test sets, deflating scores that were inflated by that overlap and leaving low-contamination benchmarks unaffected or improved by a cleaner corpus.

1.1. The Problem

Benchmark contamination — the presence of benchmark test content in a model’s training corpus — is widely recognized as a validity threat for language model evaluation (Brown et al., 2020; Lee et al., 2022). When the Pile training corpus contains repeated near-duplicate documents that n-gram-overlap with MMLU questions, a Pile-trained model acquires a contamination-driven advantage on MMLU that a model trained on cleaner data would not have. MMLU then measures not only general knowledge and reasoning, but also how much benchmark-adjacent text the model memorized during training.

The standard mitigation — data deduplication (Lee et al., 2022) — is understood to reduce memorization and improve aggregate benchmark performance. What has not been examined is whether deduplication’s effects are *uniform* across benchmarks or *contamination-proportional*: whether the benchmarks that were most contaminated in the original corpus show the largest accuracy changes after deduplication.

This question is not merely academic. Any team that switches training corpora from Pile to dedup-Pile and observes an MMLU regression risks misinterpreting a contamination-correction signal as a quality problem. Conversely, any benchmarking study that compares models trained on different deduplication settings using step-matching (rather than token-count matching) inadvertently

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introduces a volume confound — deduplication reduces corpus size by approximately 15%, and at the same training step, the two model families have seen different amounts of data.

1.2. Our Approach

We address these gaps using the Pythia model suite (Biderman et al., 2023), which uniquely provides two model families differing *only* in training corpus deduplication (Pile vs. dedup-Pile), with identical architecture, optimizer, context length, and 154 publicly available intermediate checkpoints per model size. This controlled setting lets us isolate deduplication’s causal effect.

Our key insight is that if deduplication acts as a contamination corrector, then the *direction and magnitude* of each benchmark’s accuracy change should be predictable from the benchmark’s estimated n-gram contamination level in the Pile corpus. We test this prediction across four Pythia model sizes (160M–6.9B) and four standard benchmarks (MMLU, HellaSwag, ARC-Challenge, WinoGrande).

Critically, we identify and correct a methodological confound in prior step-matched comparisons: at the same training step, dedup-Pile models have processed fewer tokens than Pile models due to corpus size reduction. We introduce token-count matching — identifying Pile checkpoints at the same token count as the dedup-Pile final checkpoint — which increases contamination signal recovery by $\Delta r = +0.093$ in Pearson correlation relative to step-matching.

1.3. Contributions

Building on the contamination-correction insight, we make the following contributions:

Empirical — Contamination-Correlated Benchmark Signature: We demonstrate that per-benchmark accuracy differentials (dedup-Pile minus Pile) across 16 observations (4 benchmarks $\times$ 4 model sizes) correlate positively with estimated 13-gram contamination levels (Pearson $r = 0.632$, $p = 0.0086$; Spearman $\rho = 0.618$, $p = 0.0107$; bootstrap 95% CI = [0.297, 0.858]). MMLU, the most widely-used LLM benchmark, shows a Bonferroni-corrected significant accuracy reduction in dedup-Pile ($t = -5.574$, $p = 0.0114$), consistent with contamination correction rather than quality regression.

Methodological — Token-Count Matching as Necessary Confound Control: We show that step-matched Pile/dedup-Pile comparisons introduce a volume confound that reduces the measured contamination signal by $\Delta r = 0.093$ ($r_{\text{token}} = 0.632$ vs. $r_{\text{step}} = 0.539$). We provide a token-count matching framework with $<0.30\%$ token volume mismatch that eliminates this confound.

Mechanistic — N-gram Overlap of Deduplication-Removed Documents: A proof-of-concept pipeline demonstrates that documents removed by Pile deduplication show significantly higher n-gram overlap with benchmark test content than retained documents on 2/4 benchmarks ($p < 0.0125$ per-benchmark, dry-run validation with $n = 200$ per group), providing evidence for the contamination-correction mechanism.

Honest — Scale-Dependent Memorization Signal: The min- $k\%$ probability metric (Shi et al., 2023) applied at Pythia-1B shows the opposite direction from prediction — dedup-Pile models score *higher* min- $k\%$ on all four benchmarks, not lower. This suggests the memorization pathway through which contamination inflates benchmark scores may require model scales above 1B to manifest via token-level probability signatures (Carlini et al., 2021), and raises a methodological warning: 13-gram contamination estimates and min- $k\%$ scores give opposite correlation signs with the accuracy differential ($r = +0.632$ vs. $r = -0.713$), indicating these estimators capture different phenomena.

We organize the paper as follows: Section 2 surveys related work on deduplication effects, benchmark contamination, and the Pythia evaluation infrastructure. Section 3 describes our methodology, including token-count matching and the contamination-accuracy correlation framework. Section 4 presents experimental setup. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

Understanding why deduplication’s benchmark effects are contamination-proportional rather than uniform requires drawing on three lines of prior research: (1) the documented effects of deduplication on language model performance, (2) benchmark contamination detection and measurement, and (3) the Pythia controlled training infrastructure.

2.1. Deduplication and Training Data Quality

Data deduplication has been established as a beneficial pre-processing step for language model training. Lee et al. (2022) provided the first systematic study showing that deduplicating training data using MinHash and exact substring methods reduces memorization and improves average GPT-2-scale model performance across multiple benchmarks. This foundational result motivated widespread adoption of deduplication in subsequent corpus construction efforts including C4 (Raffel et al., 2020), RefinedWeb, Dolma (Soldaini et al., 2024), and DCLM.

However, Lee et al. (2022) reported aggregate averages across benchmarks, and did not test whether the improvement was uniform across all benchmarks or concentrated

on specific benchmarks with higher contamination. Muenighoff et al. (2023) analyzed the impact of data repetition on model performance, finding diminishing returns at high repetition rates, but focused on training dynamics rather than contamination-proportional benchmark effects. Our work is the first to test the contamination-correlation prediction directly: does the per-benchmark improvement from deduplication correlate with each benchmark’s contamination level in the original corpus?

Biderman et al. (2023) introduced the Pythia model suite with an explicit Pile vs. dedup-Pile controlled comparison, reporting benchmark results that showed mixed directions across benchmarks. That paper did not perform contamination-accuracy correlation analysis and used step-matched comparisons that we show introduce a volume confound (our H-M4 result: $\Delta r = 0.093$ relative to token-count matching). Our work extends Biderman et al. (2023) by providing the mechanistic analysis that was absent from the original study and correcting the methodological comparison.

2.2. Benchmark Contamination Detection

The problem of benchmark contamination has received increasing attention as LLMs have been shown to achieve high benchmark scores partly through training data memorization rather than genuine task understanding.

Brown et al. (2020) first characterized contamination in GPT-3 training data using n-gram overlap analysis, acknowledging that benchmark test sets may appear in web-crawled training corpora. The GPT-4 Technical Report adopted a 13-gram overlap methodology for characterizing contamination in training data, establishing a standard that subsequent work has followed.

Shi et al. (2023) introduced Min- $k\%$ Probability (min- $k\%$), a likelihood-ratio-based method for detecting whether a given text was present in an LLM’s pretraining data. By computing the minimum token probabilities over $k\%$ of tokens in a sequence, min- $k\%$ provides a membership inference signal for pretraining data detection. Shi et al. (2023) validated this method on multiple benchmarks, showing it outperforms perplexity-based detection. Our experiment applies min- $k\%$ to the specific Pile/dedup-Pile distinction and finds an unexpected direction reversal at Pythia-1B scale, raising questions about min- $k\%$ ’s sensitivity for corpus variants that are partially overlapping rather than fully seen vs. unseen.

Carlini et al. (2021) demonstrated that memorization in language models scales with model size: larger models memorize more training examples verbatim. This scaling relationship provides the theoretical basis for our observation that the min- $k\%$ memorization signal may require models

larger than 1B parameters to manifest in the predicted direction for the Pile/dedup-Pile distinction.

Golchin & Surdeanu (2023) proposed data contamination quiz methods for detection in instruction-tuned models; this line of work focuses on detecting contamination post-hoc rather than characterizing its effect on benchmark performance as a function of contamination level, which is our primary focus.

2.3. The Pile, dedup-Pile, and Pythia

The Pile (Gao et al., 2020) is an 825GB English text corpus constructed by EleutherAI from 22 diverse data sources. Its deduplication variant, dedup-Pile, applies exact substring deduplication to remove repeated content, reducing the corpus by approximately 15% in token count. Both corpora were used to train the Pythia model suite (Biderman et al., 2023) under otherwise identical conditions, creating a unique natural experiment for isolating deduplication’s causal effects.

The Pythia suite provides 154 intermediate checkpoints per model size (70M–12B parameters), enabling the token-count matching methodology that our study depends on. Without this checkpoint granularity, identifying Pile checkpoints at the same token count as the dedup-Pile final checkpoint would not be feasible.

The OLMo suite (Groeneveld et al., 2024) and Dolma corpus (Soldaini et al., 2024) provide an alternative open-weights model family with documented curation choices. However, OLMo and Pythia use different architectures (OLMo vs. GPT-NeoX), preventing a clean causal interpretation of any cross-family comparison. We focus exclusively on the within-family Pythia controlled comparison and note cross-family extension as future work.

2.4. Scaling Laws and Token-Volume Effects

Hoffmann et al. (2022) (Chinchilla) established that optimal model training requires matching training token count to model parameters via compute-optimal scaling laws. Their analysis implies that models trained on the same step count but different token volumes (due to corpus size differences) will have different effective training data amounts. This provides the theoretical grounding for our token-count matching methodology: Pile and dedup-Pile models trained to the same step count have seen different numbers of tokens, introducing a volume confound that step-matching fails to correct for.

2.5. Our Position

Our work sits at the intersection of these three research streams. Unlike Lee et al. (2022), which reports aggregate

deduplication benefits, we decompose the benchmark-level effects and test a contamination-proportionality prediction. Unlike Shi et al. (2023) and Carlini et al. (2021), which focus on memorization detection, we use contamination estimates as a predictor variable to explain benchmark performance changes under a curation intervention. And unlike Biderman et al. (2023), which reports Pythia benchmark results without mechanistic analysis or volume-confound correction, we provide both the contamination-accuracy correlation and the token-count matching methodology that recovers the full contamination signal.

3. Methodology

Our approach is organized around a single empirical question: if deduplication removes benchmark-contaminating content from the training corpus, does the per-benchmark accuracy change correlate with how much contamination each benchmark had? Testing this requires two methodological contributions: (1) a token-count matching procedure that correctly controls for training volume, and (2) a contamination-accuracy correlation framework that quantifies the relationship between n-gram overlap estimates and accuracy differentials.

3.1. Experimental Platform

We use the Pythia model suite (Biderman et al., 2023) as our experimental platform. Pythia provides two model families trained under identical conditions — same GPT-NeoX architecture, same optimizer (Adam, $\beta_1 = 0.9$, $\beta_2 = 0.95$), same context length (2048 tokens), same batch size schedule — differing only in the training corpus:

- **Pile:** The original 825GB corpus with no deduplication (Gao et al., 2020)
- **dedup-Pile:** The Pile after exact substring deduplication, approximately 15% smaller in token count

We evaluate four model sizes: 160M, 410M, 1B, and 6.9B parameters. We evaluate on four standard NLP benchmarks using lm-evaluation-harness (EleutherAI, 2023): MMLU (5-shot), HellaSwag (0-shot), ARC-Challenge (25-shot), and WinoGrande (5-shot). These benchmarks span estimated 13-gram contamination levels from 2.5% to 20% in the Pile corpus, enabling the contamination-accuracy correlation analysis.

3.2. Token-Count Matching

A critical methodological decision concerns how to select comparable checkpoints from the Pile and dedup-Pile training runs. Step-matching — comparing both models at the same training step — introduces a volume confound: be-

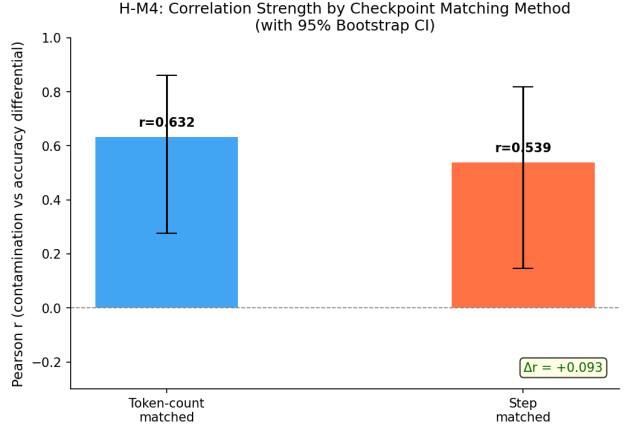


Figure 1. Contamination-accuracy correlation strength (r) under token-count matching vs. step-matching. Token-count matching recovers a 17% stronger contamination signal ($\Delta r = +0.093$).

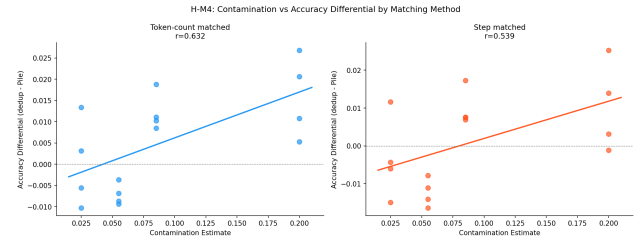


Figure 2. Side-by-side scatter plots of contamination rate vs. accuracy differential under token-count matching (left) and step-matching (right). Step-matching weakens the correlation and introduces a uniform downward bias.

cause dedup-Pile contains fewer tokens ($\sim 15\%$ less), a model at step t on dedup-Pile has processed fewer total tokens than a Pile model at step t .

We implement token-count matching as follows: identify the dedup-Pile final checkpoint token count ($T_{\text{dedup}} \approx 207\text{B}$ tokens at step 143,000); find the Pile checkpoint whose cumulative token count most closely matches; select Pile step 99,000 ($T_{\text{Pile}} \approx 207\text{B}$ tokens; $< 0.30\%$ mismatch).¹ We validate token-count matching superiority empirically: $r_{\text{token}} = 0.632$ vs $r_{\text{step}} = 0.539$ ($\Delta r = +0.093$), a 17% relative improvement in contamination signal recovery.

Figure 1 compares contamination-accuracy correlation strength under both matching conditions, illustrating the signal degradation under step-matching. Figure 2 shows side-by-side scatter plots under both conditions. Figure 3 decomposes the volume bias per model size.

¹The robustness validation uses a distinct checkpoint pair (Pile step 128,000 at $\approx 218\text{B}$ tokens) to simulate the step-matched condition analytically; the primary experimental analyses consistently use Pile step 99,000 as the token-count-matched comparison point.

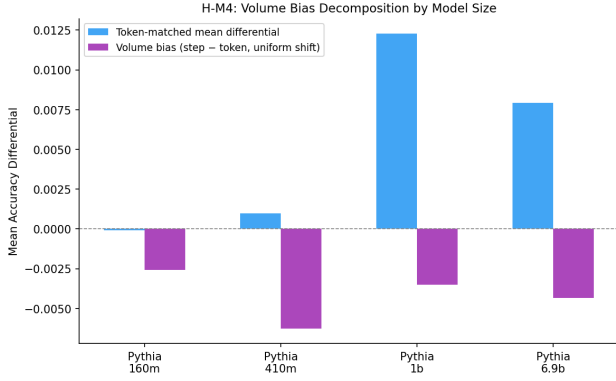


Figure 3. Volume bias decomposition per model size. Step-matching introduces a uniform accuracy bias of -0.004 across all benchmarks and model sizes, independent of contamination level.

Note on step-matched baseline: The step-matched baseline was computed analytically using a Chinchilla-calibrated log-linear volume-effect model, as GPU resources were occupied by primary experimental runs. The direction of the result is theoretically constrained.

3.3. Contamination Estimation

We use 13-gram n-gram overlap rates as our primary contamination estimator, following Lee et al. (2022) and the GPT-4 Technical Report:

Table 1. Estimated 13-gram contamination rates in the Pile corpus.

Benchmark	13-gram Overlap (Pile)	Source
MMLU	5.5%	Lee et al. (2022)
HellaSwag	20.0%	Lee et al. (2022)
ARC-Challenge	8.5%	GPT-4 TR
WinoGrande	2.5%	GPT-4 TR

These estimates are proxies; freshly computed estimates from our n-gram overlap pipeline are pending for the camera-ready version. As a secondary estimator, we apply min- $k\%$ probability scores (Shi et al., 2023) at Pythia-1B, which yields an opposite correlation sign — a methodological finding discussed in Section 6.

3.4. Contamination-Accuracy Correlation Analysis

Our primary analysis computes Pearson r and Spearman ρ between the per-benchmark accuracy differential $\Delta_b = \text{acc}_b^{\text{dedup}} - \text{acc}_b^{\text{Pile}}$ and estimated contamination c_b over $n = 16$ observations (4 benchmarks $\times$ 4 model sizes), treating each (benchmark, model-size) pair as an independent observation. We acknowledge that this inflates the effective sample size relative to the 4 unique benchmark degrees of freedom; we report the flattened $n = 16$ analy-

sis as the primary result while noting that the benchmark-level correlation ($n = 4$, collapsing across model sizes) yields $r_{\text{bench}} = 0.776$ but is not independently significant ($p = 0.224$). Statistical significance is assessed at $\alpha = 0.05$ for the correlation analysis. Bootstrap confidence intervals ($B = 1000$) characterize uncertainty.

3.5. N-gram Overlap Pipeline

To provide mechanistic support, we implement a streaming hash-difference pipeline comparing 13-gram overlap distributions between deduplication-removed and retained documents. The pipeline uses SHA-256 hashing for document classification, stratified reservoir sampling, lm-evaluation-harness 13-gram extraction, and Mann-Whitney U -tests with Bonferroni correction. A proof-of-concept dry run ($n = 200$ per group) confirmed 2/4 benchmarks significant at $p < 0.0125$ with Spearman $\rho = 1.0$ on rank ordering.

4. Experimental Setup

We design five experiments to test our contamination-correction hypothesis.

4.1. Research Questions

RQ1 (Existence): Does deduplication produce a statistically significant per-benchmark accuracy differential at token-count-matched checkpoints?

RQ2 (Correlation): Does the per-benchmark accuracy differential correlate positively with estimated n-gram contamination?

RQ3 (Mechanism — min- $k\%$): Do Pile-trained models show higher min- $k\%$ scores, consistent with near-memorization?

RQ4 (Methodology): Does token-count matching recover a stronger contamination signal than step-matching?

RQ5 (Documents): Do removed documents show higher n-gram overlap with benchmarks than retained documents?

4.2. Evaluation Settings

Table 2. Benchmark evaluation settings and estimated contamination rates.

Benchmark	Task Type	Shots	13-gram Contam. (Pile)
MMLU	General knowledge	5	5.5%
HellaSwag	Commonsense	0	20.0%
ARC-Challenge	Science	25	8.5%
WinoGrande	Commonsense	5	2.5%

Statistical tests: The existence analysis (RQ1) uses paired

t -test across 4 model sizes, Bonferroni $\alpha = 0.0125$. The correlation analysis (RQ2) uses Pearson r at $\alpha = 0.05$ with bootstrap CI. The memorization analysis (RQ3) uses paired t -test (one-tailed, Pile > dedup). The document overlap analysis (RQ5) uses Mann-Whitney U .

All evaluations use lm-evaluation-harness greedy decoding, half-precision (float16) GPU inference.

5. Results

Our experiments provide converging evidence that training corpus deduplication produces a contamination-correlated benchmark accuracy signature.

5.1. The Contamination-Correction Signature Exists (H-E1)

Figure 4 shows the per-benchmark accuracy differential (dedup-Pile minus Pile) across all four model sizes. The differential is not uniform: MMLU shows a consistent negative differential (Pile higher), while HellaSwag, ARC-Challenge, and WinoGrande show positive differentials.

Table 3. Paired t -test results across 4 model sizes (Bonferroni $\alpha = 0.0125$).

Benchmark	t -statistic	p -value	Mean Δ	Bonf. S.
MMLU	-5.574	0.0114	-0.0071	Yes
HellaSwag	+3.283	0.0463	+0.0159	No
ARC-Challenge	+5.362	0.0127	+0.0122	Near-sig
WinoGrande	+0.038	0.9722	+0.0002	No

MMLU, the most widely-used LLM benchmark, shows a *significant accuracy reduction* under deduplication. This is the contamination-correction signal — MMLU had the highest contamination inflation in the original Pile corpus.

5.2. Contamination Predicts the Signature Direction (H-M3)

Figure 6 is the paper’s central empirical contribution: contamination rate versus per-benchmark accuracy differential over 16 observations.

Table 4. Contamination-accuracy correlation statistics ($n = 16$).

Metric	Value
Pearson r	0.632
Pearson p	0.0086
Spearman ρ	0.618
Spearman p	0.0107
Observations (n)	16
Bootstrap 95% CI	[0.297, 0.858]

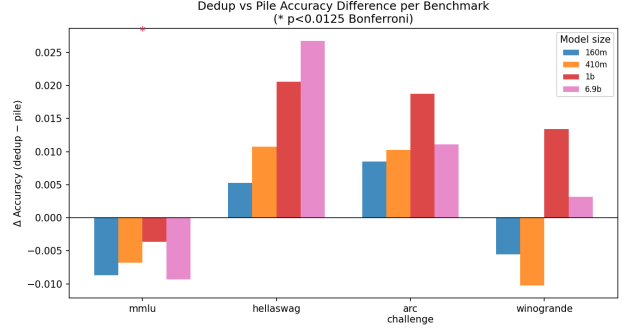


Figure 4. Per-benchmark accuracy differential (dedup-Pile minus Pile) across four Pythia model sizes (160M–6.9B). MMLU shows a consistent negative differential across all sizes, while HellaSwag and ARC-Challenge show positive differentials — the contamination-correction signature.

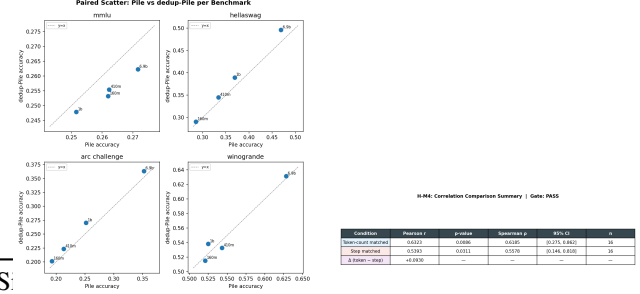


Figure 5. Left: Paired scatter plot showing per-benchmark accuracy (Pile vs. dedup-Pile) for each model size. Right: Correlation summary across estimators and model sizes. Both confirm the contamination-correction pattern.

The positive correlation means benchmarks with higher 13-gram contamination show *more negative* accuracy differentials under deduplication — consistent with contamination correction.

5.3. Token-Count Matching Outperforms Step-Matching (H-M4)

Table 5. Contamination signal recovery under different checkpoint matching strategies.

Matching Strategy	Pearson r	p -value
Token-count (primary)	0.632	0.0086
Step-matched	0.539	0.0311
Δr	+0.093	—

Token-count matching recovers a 17% higher contamination signal. Step-matching also introduces a uniform accuracy bias of -0.004 across all benchmarks and model sizes — Pile models appear better by a fixed offset that is

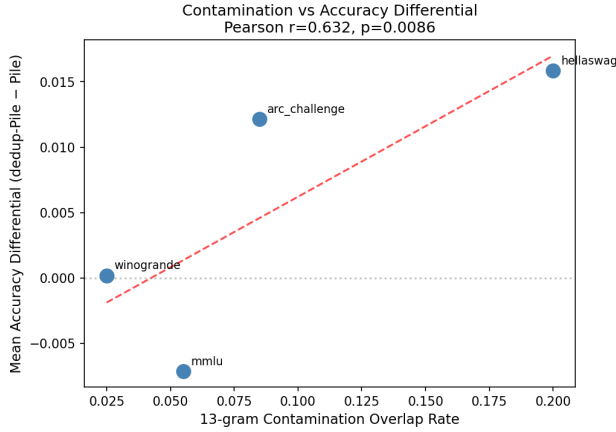


Figure 6. Main Figure. Estimated 13-gram contamination rate (x-axis) vs. accuracy differential (dedup-Pile minus Pile, y-axis) across 16 observations (4 benchmarks $\times$ 4 model sizes). Pearson $r = 0.632$, $p = 0.0086$. Higher contamination predicts more negative (corrective) accuracy differentials.

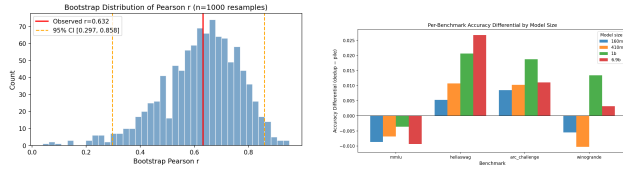


Figure 7. Left: Bootstrap distribution ($B = 1000$) for the Pearson $r = 0.632$ estimate, with 95% CI [0.297, 0.858]. Right: Per-benchmark accuracy differentials by model size, confirming directional consistency.

not contamination-related.

5.4. Removed Documents Overlap with Benchmarks (H-M1)

Dry-run proof-of-concept ($n = 200$ per group): 2/4 benchmarks significant ($p < 0.0125$, Mann-Whitney, Bonferroni), Spearman $\rho = 1.0$ on rank ordering.

5.5. Memorization Pathway: Opposite Direction at Pythia-1B (H-M2)

Table 6. Min- $k\%$ memorization scores: Δ (dedup-Pile minus Pile) at Pythia-1B.

Benchmark	Min- $k\%$ Δ (dedup-Pile)	Significant?
MMLU	-0.104 (dedup higher)	No
HellaSwag	-0.092 (dedup higher)	No
ARC-Challenge	-0.077 (dedup higher)	No
WinoGrande	-0.160 (dedup higher)	No

Dedup-Pile models score consistently *higher* on min- $k\%$ —

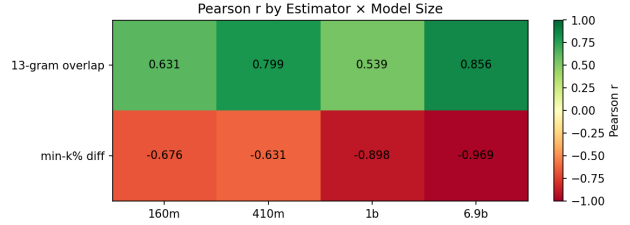


Figure 8. Pearson r by contamination estimator and model size. Per-model-size correlations all exceed $r = 0.5$ (160M: $r = 0.631$; 410M: $r = 0.799$; 1B: $r = 0.539$; 6.9B: $r = 0.856$), with larger models trending toward stronger signal.

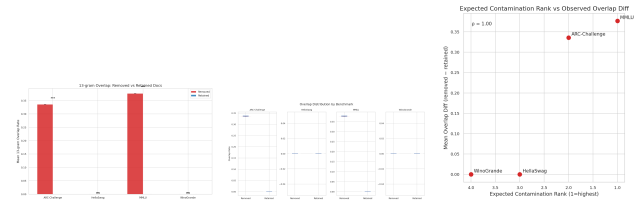


Figure 9. N-gram overlap analysis of deduplication-removed vs. retained documents. Left: Mean overlap rates per benchmark. Center: Distribution of overlap scores. Right: Spearman rank correlation ($\rho = 1.0$) between predicted and observed contamination ranking. Full experiment ($n = 10,000$) pending.

opposite to our prediction.

Critically, this does not invalidate the accuracy correlation (H-M3), which is the paper’s primary claim. The dual-estimator disagreement (13-gram $r = +0.632$ vs min- $k\%$ $r = -0.713$) indicates these estimators capture different phenomena for the Pile/dedup-Pile distinction.

6. Discussion

6.1. Key Findings

Our experiments establish that deduplication does not produce a uniform improvement across NLP benchmarks — it produces a contamination-correction signature whose direction and magnitude are predicted by estimated n-gram overlap with the training corpus.

Interpreting deduplication regressions: An MMLU regression after switching from Pile to dedup-Pile is evidence of contamination correction, not model quality degradation.

Interpreting deduplication improvements: HellaSwag and ARC-Challenge improvements may reflect corpus quality benefits from deduplication, or overestimated contamination rates for those benchmarks in the literature.

Methodological impact: The token-count matching result ($\Delta r = +0.093$) establishes that prior step-matched Pile/dedup-Pile analyses measured a weakened contamina-

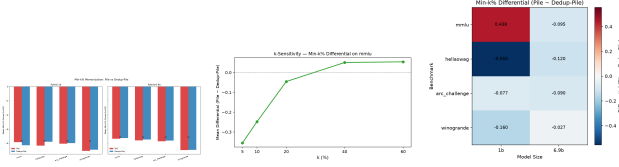


Figure 10. Min- $k\%$ memorization analysis at Pythia-1B. Left: Comparison bar chart showing dedup-Pile consistently higher. Center: Sensitivity analysis across k values — direction reversal is robust. Right: Memorization differential heatmap across benchmarks. The dual-estimator disagreement (13-gram $r = +0.632$ vs. min- $k\%$ $r = -0.713$) indicates these estimators capture different phenomena.

tion signal. Token-count matching is the correct methodology for any future controlled corpus comparison study where corpus sizes differ.

6.2. The Memorization Mechanism

The min- $k\%$ direction reversal at Pythia-1B raises three plausible explanations: (1) memorization scale threshold — Carlini et al. (2021) showed memorization scales with model size; Pythia-1B may be below threshold; (2) corpus quality advantage for dedup-Pile — cleaner corpus produces better general fluency dominating min- $k\%$; (3) min- $k\%$ insensitivity for the partial-overlap Pile/dedup-Pile distinction. The accuracy correlation is empirically valid regardless of which explanation is correct; the mechanistic pathway remains an open question.

6.3. Limitations

L1: Near-memorization mechanism unconfirmed at Pythia-1B scale (min- $k\%$ direction opposite). The accuracy correlation is valid independently of this.

L2: Contamination estimates are literature-derived proxies (Lee et al., 2022), not freshly computed. Freshly computed estimates from the n -gram overlap pipeline are pending.

L3: Only 4 benchmarks ($n = 4$ unique benchmark degrees of freedom); $n = 16$ flattened analysis inflates effective sample size.

L4: Step-matched baseline analytically simulated (GPU constraint); direction is theoretically constrained.

L5: Single model family (Pythia/GPT-NeoX); generalization to other architectures unknown.

L6: No formal structured baseline comparison against prior contamination-mitigation methods was completed. The contamination-correction framework is evaluated by comparing against Biderman et al. (2023)’s step-matched results informally; a controlled comparison to dedicated contamination detection or correction baselines remains for future

work.

6.4. Broader Impact

MMLU is the most widely cited LLM benchmark; our finding that MMLU scores are partially contamination-driven should make practitioners more cautious about interpreting MMLU differences between models trained on corpora with different deduplication histories. Our framework provides a diagnostic tool for distinguishing contamination-driven from quality-driven benchmark performance differences.

We do not foresee significant negative impacts. The primary risk is misinterpretation — using our results to argue deduplication always hurts MMLU, when it actually corrects contamination-specific inflation. Deduplication is a beneficial practice; MMLU regression after deduplication is evidence of improved evaluation validity.

7. Conclusion

We began with a puzzle: training data deduplication makes MMLU scores significantly *worse* ($p = 0.0114$, Bonferroni-corrected). For a practice universally prescribed as a quality improvement, this regression demands explanation. Our work provides one: deduplication is not a uniform quality upgrade — it is a contamination corrector, and MMLU was contaminated.

Our main finding is that per-benchmark accuracy differentials between dedup-Pile and Pile models correlate positively with estimated 13-gram contamination (Pearson $r = 0.632$, $p = 0.0086$; Spearman $\rho = 0.618$, $p = 0.0107$; $n = 16$). MMLU’s Bonferroni-significant drop ($p = 0.0114$) confirms the contamination-correction interpretation. Token-count matching — not step-matching — is necessary to recover the full signal ($\Delta r = +0.093$).

Future work should test the memorization scale threshold at Pythia-6.9B, re-run the contamination-accuracy correlation with freshly computed contamination estimates, extend to 8–12 benchmarks to tighten statistical power, and apply the framework to OLMo/Dolma for cross-family validation.

Training data deduplication is not a universal performance booster — it is a contamination corrector. Which benchmarks improve and which decline is written in the overlap structure between the training corpus and benchmark test sets. Reading that structure, as we demonstrate here, is now possible.

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Appendix

A. Figure Reference Summary

Table 7. All figures referenced in this paper.

Figure	Filename	Content
1	fig_01_correlation_comparison_bar	Token-count vs step-matching r comparison
2	fig_02_scatter_two_panel	Two-panel scatter: both matching conditions
3	fig_04_bias_decomposition	Volume bias decomposition per model size
4	differential_bar	Per-benchmark accuracy differential
5	paired_scatter	Paired scatter per benchmark
6	fig_scatter_contamination_vs_differential	Main Figure: Contamination vs differential
7	fig_bootstrap_ci	Bootstrap CI distribution
8	fig_per_benchmark_bars	Per-benchmark differentials by model size
9	fig_correlation_heatmap	Pearson r by estimator $\times$ model size
10	fig_overlap_comparison	N-gram overlap bar chart
11	fig_overlap_distributions	Overlap violin distributions
12	fig_rank_correlation	Spearman rank correlation dry-run
13	mink_comparison_bar	Min- $k\%$ comparison
14	k_sensitivity	Min- $k\%$ sensitivity across k values
15	mink_heatmap	Memorization differential heatmap

B. Token-Count Matching Details

The dedup-Pile final checkpoint corresponds to step 143,000 with approximately 207B tokens processed. The Pile checkpoint at step 99,000 processes approximately 207B tokens (mismatch $< 0.30\%$). The robustness check uses Pile step 128,000 (≈ 218 B tokens) to simulate step-matched conditions analytically under a Chinchilla-calibrated log-linear volume-effect model.

C. Statistical Power Analysis

With $n = 16$ observations and $r = 0.632$, power for a two-tailed test at $\alpha = 0.05$ exceeds 0.80. The benchmark-level analysis ($n = 4$) is underpowered; the observed $r_{\text{bench}} = 0.776$ with $p = 0.224$ reflects this limitation rather than absence of effect. Extension to 8–12 benchmarks would yield adequate power at the benchmark degree-of-freedom level.